

Homework 3 — Software Requirements Specification

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Electromechanical Maintenance Management System.

This document describes the system's expected functionalities, user interactions, and operational constraints. It serves as a reference for the development team and stakeholders throughout the software development process, including system design, implementation, and testing.

The SRS aims to establish a clear understanding of what the system must accomplish and ensure that its development addresses the maintenance management needs identified in the project.

This document presents an initial proposal of software requirements. The proposed scope and requirements are subject to review, validation, and changes following feedback from the client, Enrique Nacimba.

1.2 Scope

The Electromechanical Maintenance Management System is a software solution designed to organize and manage maintenance operations in electromechanical workshops and companies.

The system will centralize information about clients, technicians, and electromechanical equipment. It will allow users to register and update records, create and assign work orders, schedule preventive maintenance, and document corrective maintenance activities.

Additionally, the system will provide access to equipment maintenance history, generate unique QR codes for equipment identification, and produce maintenance reports and alerts for upcoming services.

The primary objective is to improve maintenance organization, facilitate access to technical information, reduce unexpected equipment downtime, and support technical and administrative decision-making.

1.3 Definitions and Acronyms

The following acronyms are used throughout this document. Project terminology is defined in Appendix A.

Term	Definition
SRS	Software Requirements Specification. A document that defines the requirements of a software system.
FR	Functional Requirement. A requirement that describes a function the system must perform.
NFR	Non-Functional Requirement. A requirement that describes a quality attribute or operational constraint of the system.

1.4 References

The following project documents provide the foundation for this Software Requirements Specification:

Binary Bosses. *HW04 – Definition: Electromechanical Maintenance Management System*. University of the Armed Forces – ESPE, 2026.

Binary Bosses. *HW05 – Requirements: Product Backlog*. University of the Armed Forces – ESPE, 2026.

2. Overall Description

2.1 Product Perspective

The Electromechanical Maintenance Management System is a software solution designed to centralize and organize maintenance operations in electromechanical workshops and companies.

Currently, maintenance information may be managed through spreadsheets, paper records, or disconnected tools, making it difficult to coordinate clients, technicians, equipment, and maintenance activities. The proposed system aims to address these limitations by providing a centralized environment for managing maintenance-related information.

The system will integrate client and equipment registration, technician management, work order assignment, maintenance scheduling, and maintenance history. It will also support QR code identification, maintenance reports, and alerts for upcoming services.

By organizing these operations within a single system, the software is expected to improve information accessibility, maintenance coordination, and operational efficiency.

2.2 Product Functions

The main functions of the Electromechanical Maintenance Management System are described below:

Client Management: Register, search, and update client information to maintain organized client records.

Equipment Management: Register, search, and update electromechanical equipment information, linking each machine to its corresponding client.

Technician Management: Register technicians and manage their availability and specializations to support maintenance assignments.

Work Order Management: Create maintenance work orders and assign them to available technicians.

Preventive Maintenance Scheduling: Schedule preventive maintenance activities and register upcoming equipment inspections.

Corrective Maintenance Records: Record equipment repairs, technical observations, and replaced parts during corrective maintenance.

Maintenance History: Search and view previous maintenance activities associated with registered equipment.

QR Code Identification: Generate unique QR codes linked to equipment records, allowing users to access equipment information by scanning them.

Maintenance Reports and Alerts: Generate maintenance summaries and notify users about upcoming scheduled services.

These functions correspond to the nine features established in the project's Product Backlog.

2.3 User Classes and Characteristics

The system is intended to support three main user roles:

Administrator

The administrator is responsible for coordinating maintenance operations and managing system information. This role includes maintaining client, technician, and equipment records, assigning work orders, organizing maintenance schedules, and reviewing maintenance reports.

Technician

The technician performs preventive and corrective maintenance activities. This role uses the system to access assigned work orders, consult equipment information and maintenance history, and record completed maintenance activities, technical observations, and replaced parts.

Client

The client is associated with the equipment registered in the system. This role is intended to access relevant information about their equipment and maintenance records, according to the access permissions established during system development.

2.4 Operating Environment

The Electromechanical Maintenance Management System is intended to operate in a digital environment that supports the registration, storage, retrieval, and management of maintenance information.

The system will require access to a data storage mechanism to maintain client, technician, equipment, work order, and maintenance records.

Its user interface should support the management of maintenance activities and the consultation of registered information. Additionally, the system must provide a mechanism for generating and scanning QR codes associated with equipment records.

The specific operating system, programming language, database management system, deployment platform, and hardware requirements have not yet been established and must be defined during the technical design phase.

2.5 Design and Implementation Considerations

The system design should maintain consistency among clients, equipment, technicians, work orders, and maintenance records.

Equipment information must remain associated with its corresponding client, while maintenance activities must be connected to the relevant equipment and work orders.

The software should also organize preventive and corrective maintenance information so that previous interventions can be consulted through the equipment maintenance history.

The implementation must support the nine features established in the Product Backlog, respecting their defined priorities.

2.6 Assumptions and Dependencies

The proposed system assumptions and operational dependencies are specified in Section 7. They will be reviewed during stakeholder validation.

3. Functional Requirements

This section describes the main functional requirements of the Electromechanical Maintenance Management System. The requirements are based on the project's Product Backlog and define the essential functions proposed for the system.

3.1 FR-01: Client Management

Description: The system shall allow users to manage client information.

Functional Requirements:

- FR-01.1: The system shall allow users to register and update client information.
- FR-01.2: The system shall allow users to search for registered clients.

Priority: High

Product Backlog Reference: PB-01

Acceptance Criteria: Users can register, search for, and update client records.

3.2 FR-02: Equipment Management

Description: The system shall allow users to manage electromechanical equipment associated with clients.

Functional Requirements:

- FR-02.1: The system shall allow users to register and update equipment information.
- FR-02.2: The system shall allow users to search for equipment and identify its associated client.

Priority: High

Product Backlog Reference: PB-02

Acceptance Criteria: Users can register, search for, and update equipment records linked to clients.

3.3 FR-03: Technician Management

Description: The system shall allow users to manage technician information.

Functional Requirements:

- FR-03.1: The system shall allow users to register technicians and their specializations.
- FR-03.2: The system shall allow users to register and view technician availability.

Priority: High

Product Backlog Reference: PB-03

Acceptance Criteria: Technician information, specializations, and availability can be registered and consulted.

3.4 FR-04: Work Order Management

Description: The system shall allow users to create and assign maintenance work orders.

Functional Requirements:

- FR-04.1: The system shall allow users to create work orders for registered equipment.
- FR-04.2: The system shall allow users to assign work orders to available technicians.

Priority: High

Product Backlog Reference: PB-04

Acceptance Criteria: Users can create work orders and assign them to available technicians.

3.5 FR-05: Preventive Maintenance Scheduling

Description: The system shall allow users to schedule preventive maintenance activities.

Functional Requirements:

- FR-05.1: The system shall allow users to schedule preventive maintenance for registered equipment.
- FR-05.2: The system shall allow users to view upcoming maintenance inspections.

Priority: High

Product Backlog Reference: PB-05

Acceptance Criteria: Users can register maintenance dates and consult scheduled inspections.

3.6 FR-06: Corrective Maintenance Records

Description: The system shall allow users to record corrective maintenance activities.

Functional Requirements:

- FR-06.1: The system shall allow users to record equipment repair details and technical observations.
- FR-06.2: The system shall allow users to register parts replaced during maintenance.

Priority: High

Product Backlog Reference: PB-06

Acceptance Criteria: Users can save repair details, technical observations, and replaced parts.

3.7 FR-07: Equipment Maintenance History

Description: The system shall allow users to consult the maintenance history of equipment.

Functional Requirements:

- FR-07.1: The system shall allow users to search for the maintenance history of registered equipment.
- FR-07.2: The system shall display previously recorded maintenance activities for the selected equipment.

Priority: High

Product Backlog Reference: PB-07

Acceptance Criteria: Users can search for and view previous maintenance records associated with equipment.

3.8 FR-08: Equipment QR Code Identification

Description: The system shall support equipment identification through QR codes.

Functional Requirements:

- FR-08.1: The system shall generate a unique QR code linked to each equipment record.
- FR-08.2: The system shall allow users to access equipment information by scanning its QR code.

Priority: Medium

Product Backlog Reference: PB-08

Acceptance Criteria: The generated QR code identifies the corresponding equipment and provides access to its information.

3.9 FR-09: Maintenance Reports and Alerts

Description: The system shall provide maintenance summaries and upcoming service alerts.

Functional Requirements:

- FR-09.1: The system shall generate summaries of recorded maintenance activities.
- FR-09.2: The system shall display upcoming scheduled maintenance alerts within the application.

Priority: Medium

Product Backlog Reference: PB-09

Acceptance Criteria: Users can view maintenance summaries and alerts for scheduled services.

The relationship between functional requirements and Product Backlog items is provided in Appendix B. These requirements remain subject to stakeholder feedback.

4. Non-Functional Requirements

This section describes the quality requirements of the Electromechanical Maintenance Management System. These requirements define the expected performance, security, usability, reliability, and maintainability of the proposed software.

4.1 NFR-01: Performance

The system shall respond to basic operations, such as registering, searching, and updating records, within 3 seconds under normal operating conditions.

Acceptance Criteria: Basic operations are completed within 3 seconds during testing with sample data.

4.2 NFR-02: Security

The system shall require users to log in with a username and password before accessing maintenance information.

Acceptance Criteria: Users can access the system with valid credentials, while invalid login attempts are rejected.

4.3 NFR-03: Usability

The system shall provide a simple and understandable user interface that allows users to navigate between its main functions.

Acceptance Criteria: Users can locate and access the main system functions through clearly labeled menus and buttons.

4.4 NFR-04: Reliability

The system shall store registered information and preserve it when the application is closed and reopened.

Acceptance Criteria: Previously saved records remain available after restarting the application.

4.5 NFR-05: Maintainability

The system shall use an organized object-oriented structure with separate classes representing its main entities.

Acceptance Criteria: The source code contains clearly identified classes with defined responsibilities, allowing individual components to be modified without unnecessarily affecting unrelated components.

5. External Interface Requirements

This section describes the external interfaces required for the Electromechanical Maintenance Management System. These interfaces define how users interact with the application and how the system communicates with the hardware and software components necessary for its operation.

5.1 User Interfaces

The system shall provide a simple and organized graphical user interface that allows users to access its main functionalities.

- UI-01: The system shall provide a main menu to access client, equipment, technician, and maintenance management functions.
- UI-02: The system shall provide forms to register and update information.
- UI-03: The system shall display registered information through organized lists and tables.
- UI-04: The system shall display clear messages to confirm completed operations or identify input errors.

5.2 Hardware Interfaces

The system shall operate on a computer with the necessary hardware resources to execute the application.

- HI-01: The system shall support standard input devices, such as a keyboard and mouse.
- HI-02: The system shall display information through a computer monitor.
- HI-03: The system shall support the use of a QR code scanning device or compatible camera for equipment identification.

5.3 Software Interfaces

The system shall interact with the software components necessary to manage and store maintenance information.

- SI-01: The system shall use a data storage mechanism to save and retrieve registered information.
- SI-02: The system shall support the generation of QR codes associated with equipment records.
- SI-03: The system shall provide access to stored maintenance information through its application interface.

5.4 Communication Interfaces

The initial proposal does not require integration with external network services. Equipment QR codes may be scanned using a compatible device, with access to associated records handled by the application as described in FR-08.

6. Constraints

This section describes the main constraints that must be considered during the development of the Electromechanical Maintenance Management System. These constraints establish the proposed technical and operational limitations of the project.

6.1 Technical Constraints

- **CON-01:** The system shall be developed using object-oriented programming principles to organize its main functionalities.
- **CON-02:** The system shall use a simple data storage mechanism to manage clients, equipment, technicians, and maintenance records.

6.2 Operational Rules

- **CON-03: Users must provide the necessary information when registering equipment and maintenance activities.**
- **CON-04: Each work order must refer to registered equipment and a registered technician.**

6.3 Project Scope Constraints

- **CON-05: The proposed scope is limited to the nine Product Backlog features; advanced extensions are excluded from the initial version.**

These constraints represent the initial proposed limitations of the system and may be adjusted based on the development team's capabilities and stakeholder feedback.

7. Assumptions and Dependencies

This section describes the assumptions and dependencies considered for the development and operation of the Electromechanical Maintenance Management System.

7.1 Assumptions

The following assumptions are considered for the proposed system:

- **AS-01:** Users are expected to provide accurate information when registering clients, equipment, technicians, and maintenance activities.

- **AS-02:** Users are expected to have basic computer skills to operate the system.
- **AS-03:** Technicians are expected to update maintenance records after completing their assigned activities.

7.2 Dependencies

The system's operation will depend on the following conditions:

- **DEP-01:** Work order assignments will depend on the availability information registered for technicians.
- **DEP-02:** Maintenance history will depend on previously recorded maintenance activities.
- **DEP-03:** QR code identification will depend on the correct association between QR codes and equipment records.
- **DEP-04:** Maintenance alerts will depend on the preventive maintenance schedules registered in the system.

These assumptions and dependencies represent the initial conditions considered for the proposed software and may be reviewed according to stakeholder feedback.

8. Appendices

This section provides supplementary information to support the understanding and organization of the Software Requirements Specification (SRS) for the Electromechanical Maintenance Management System.

Appendix A: Glossary

The following terms describe the main concepts used in the proposed system.

Term	Definition
Client	A person or organization that requests maintenance services for electromechanical equipment.
Equipment	An electromechanical machine or device registered in the system.
Technician	A professional responsible for performing maintenance activities.
Work Order	A record describing a maintenance activity assigned to a technician.
Preventive Maintenance	Planned maintenance performed to reduce the probability of equipment failure.
Corrective Maintenance	Maintenance performed to repair equipment after a failure occurs.
Maintenance History	A collection of previous maintenance records associated with equipment.
QR Code	A machine-readable code used to identify equipment and access associated information.

Maintenance Report	A summary of maintenance information recorded in the system.
Maintenance Alert	A notification about an upcoming scheduled maintenance activity.

Appendix B: Requirements Traceability Matrix

The Requirements Traceability Matrix establishes the relationship between the functional requirements defined in this document and the features identified in the Product Backlog.

Requirement ID	Functional Requirement	Product Backlog	Priority
FR-01	Client Management	PB-01	High
FR-02	Equipment Management	PB-02	High
FR-03	Technician Management	PB-03	High
FR-04	Work Order Management	PB-04	High
FR-05	Preventive Maintenance Scheduling	PB-05	High
FR-06	Corrective Maintenance Records	PB-06	High
FR-07	Equipment Maintenance History	PB-07	High
FR-08	Equipment QR Code Identification	PB-08	Medium
FR-09	Maintenance Reports and Alerts	PB-09	Medium

This matrix ensures that each functional requirement corresponds to a feature established in the Product Backlog. It also facilitates future requirement reviews, modifications, and validation with stakeholders.

Appendix C: Requirement Validation

The requirements presented in this document constitute Version 1.0 of the initial proposal for the Electromechanical Maintenance Management System.

The proposed requirements will be reviewed with the client, Enrique Nacimba, to determine whether the system addresses his maintenance management needs.

Any changes requested will be evaluated and incorporated into future versions of the SRS when appropriate.

The final requirements will be established after stakeholder review and validation.